

Generated MCQ Quiz

Q1. Which of the following is classified as a passive transducer according to the study material?

- A. Piezo-electric type accelerometer
- B. Capacitive transducer
- C. Smart accelerometer
- D. Vibration sensor

Correct Answer: B

Explanation: According to the study material, passive transducers derive power from an auxiliary external power source, and examples include resistive, inductive, and capacitive transducers. Piezo-electric accelerometers and smart accelerometers are classified as active transducers.

Q2. According to the study material, which type of thermistor is generally used as a fuse?

- A. Negative Temperature Coefficient (NTC) thermistor
- B. Positive Temperature Coefficient (PTC) thermistor
- C. Linear-response thermistor
- D. High precision epoxy thermistor

Correct Answer: B

Explanation: The study material states that a PTC (Positive Temperature Coefficient) thermistor, where resistance increases as temperature increases, is generally used as a fuse.

Q3. According to the Instrument Society of America (ISA) definition provided in the text, how is the output of a sensor/transducer finally defined?

- A. As an Electrical Quantity
- B. As a physical displacement
- C. As an auxiliary pressure wave
- D. As a mechanical strain

Correct Answer: A

Explanation: The Instrument Society of America defines a sensor/transducer as a device which provides a usable output in response to a specified measurand, where the output is defined as an Electrical Quantity (finally).

Q4. According to the classifications in the text, an Intelligent Sensor is composed of a Smart Sensor integrated with which of the following?

- A. An Inverse Transducer

- B. A Digital Processor (DPU)
- C. An auxiliary analog potentiometer
- D. A Piezo-electric accelerometer

Correct Answer: B

Explanation: The study material defines an Intelligent Sensor as a 'Smart sensor + Digital Processor (DPU)'.

Q5. What pure metal materials are typically used to construct the wire in a Resistance Temperature Detector (RTD)?

- A. Iron, aluminum, or nickel
- B. Silicon, epoxy, or glass
- C. Platinum (Pt), nickel (Ni), or copper (Cu)
- D. Zinc, brass, or platinum

Correct Answer: C

Explanation: The study material states that the RTD wire is a pure material, typically platinum (Pt), nickel (Ni), or copper (Cu).

Q6. Why is a potentiometer (POT) classified as a passive transducer according to the study material?

- A. It converts physical output into an analog voltage without drawing power.
- B. It needs an external power source for its operation.
- C. It utilizes self-generating piezo-resistive elements.
- D. It can only work with AC excitation.

Correct Answer: B

Explanation: The text explicitly states: 'The POT is a passive transducer since it needs an external power source for its operation.'

Q7. What is the state of the output voltage of the secondary windings s1 and s2 in an RVDT at its primary null position?

- A. The output voltages are equal and in phase, producing maximum voltage.
- B. The output voltages are unequal and out of phase.
- C. The output voltages are equal and in opposition, hence the net output is zero.
- D. The output voltages bypass the secondary windings entirely.

Correct Answer: C

Explanation: According to the text, 'At the primary null position of the core, The output voltage of the secondary windings s1 and s2 are equal and in opposition, hence net output zero.'

Q8. What is an inverse transducer typically known as, according to the study material?

- A. An Actuator
- B. A Smart Sensor
- C. A Strain Gauge
- D. A Thermocouple

Correct Answer: A

Explanation: The text states that an 'Inverse Transducer: Converts electrical signal to physical quantity, typically, known as "Actuator".'

Q9. For a capacitive transducer using a change in the Area of Plates for linear displacement, what is the suitable measurement range and accuracy specified in the text?

- A. 1mm to several cms, with 1% accuracy
- B. 1mm to 10mm, with accuracy as high as 0.005%
- C. 10mm to 100mm, with 0.05% accuracy
- D. 0.1mm to 1mm, with 0.1% accuracy

Correct Answer: B

Explanation: The text states that this type of capacitor is 'suitable for measurement of linear displacements ranging from 1mm -10mm. Accuracy is as high as 0.005%.'

Q10. How does the output of an Analog Transducer differ from a Digital Transducer according to the study material?

- A. Analog output is a continuous function of 'amplitude and time', while digital output is in the form of pulses or a discrete function.
- B. Analog output is in the form of pulses, while digital output is a continuous function of 'amplitude and time'.
- C. Analog output does not require an auxiliary power source, while digital output always does.
- D. Analog output only measures physical displacement, while digital output only measures temperature.

Correct Answer: A

Explanation: According to the study material, an Analog Transducer converts an input quantity into an output that is a continuous function of 'amplitude and time', while a Digital Transducer converts input into an electrical output in the form of pulses (quasidigital output) or a discrete function of 'amplitude and time'.